

# gemstone-rs Feature Map

Crates, examples, docs, and gemstone-py parity references



Generated from the Markdown sources in `gemstone-rs/docs`.

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# gemstone-rs Feature Map

This map is the Rust equivalent of `gemstone-examples plan3-map` in `gemstone-py`: it ties each feature stream to the crates, examples, docs, and Python reference point that inspired it.

Run it from the CLI:

```
gemstone-rs hello
gemstone-rs compare gemstone-py
gemstone-rs compare gemstone-py --gaps
gemstone-rs examples map
gemstone-rs examples map --json
gemstone-rs examples scaffold quickstart ./gemstone-rs-quickstart
gemstone-rs examples scaffold codegen_workflow ./gemstone-rs-codegen-workflow
gemstone-rs examples scaffold profile_codegen_workflow ./gemstone-rs-profile-codegen
gemstone-rs examples scaffold generated_wrapper_app ./gemstone-rs-generated-wrapper
gemstone-rs examples scaffold axum_service ./gemstone-rs-axum-service
```

`hello` is the no-live sanity check that mirrors `gemstone-examples hello`. `compare gemstone-py` prints a compact version of the Python/Rust comparison guide. `compare gemstone-py --gaps` prints the remaining parity gaps with next actions and verification commands. `examples scaffold` creates standalone Cargo projects from installed templates for quickstart, browser, BridgeRoot mapping, derive mapping, generated wrappers, live discovery, profile-driven codegen, standard HTTP, and Axum workflows. JSON output is intended for CI, docs checks, and editor tooling.

## Streams

| Stream | Feature                        | Rust surface                                                         | Examples                                                                                     | Docs                                                                          | gemstone                                                   |
|--------|--------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------|
| 1      | Runtime and GCI loading        | <code>gemstone-gci</code> ,<br><code>gemstone-rs::Config</code>      | <code>hello</code> , <code>hello_gemstone</code> ,<br><code>quickstart</code>                | <code>docs/setup-guide.md</code> ,<br><code>docs/performance-safety.md</code> | <code>gemstone-native</code>                               |
| 2      | Safe sessions and transactions | <code>gemstone-rs::Session</code> ,<br><code>TransactionGuard</code> | <code>quickstart</code> ,<br><code>transactions</code> ,<br><code>live_smoke_cookbook</code> | <code>docs/user-manual.md</code> ,<br><code>docs/cookbook.md</code>           | <code>GemstoneSession</code> ,<br><code>Transaction</code> |

| Stream | Feature                   | Rust surface                                                                                    | Examples                                                                                                                                                                  | Docs                                                                       | gemstone                              |
|--------|---------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------|
|        |                           |                                                                                                 |                                                                                                                                                                           |                                                                            |                                       |
| 3      | Browser and inspection    | <code>gemstone-rs::browser</code> , CLI <code>browse</code> , <code>gemstone-rs-explorer</code> | <code>browser</code> , <code>tooling/cli-browser-walkthrough.md</code>                                                                                                    | <code>docs/user-manual.md</code> , <code>docs/explorer.md</code>           | <code>gemstone-python-database</code> |
| 4      | OOP and value handling    | <code>Oop</code> , <code>Value</code> , export-set helpers                                      | <code>oop_values</code>                                                                                                                                                   | <code>docs/user-manual.md</code> , <code>docs/performance-safety.md</code> | Managed typed arrays                  |
| 5      | BridgeRoot object mapping | <code>gemstone-rs::bridge</code> , <code>gemstone-rs-macros</code>                              | <code>bridge_root_mapping</code> , <code>derive_mapping</code> , <code>generated_mapping_app</code>                                                                       | <code>docs/object-mapping.md</code> , <code>docs/cookbook.md</code>        | Small Persistent examples             |
| 6      | Typed codegen             | <code>gemstone-rs::codegen</code> , CLI <code>codegen</code> and <code>profile</code>           | <code>codegen_preview</code> , <code>codegen_workflow</code> , <code>codegen_discover</code> , <code>profile_codegen_workflow</code> , <code>generated_wrapper_app</code> | <code>docs/codegen.md</code> , <code>docs/profile-schema.md</code>         | <code>gemstone-typed-codegen</code>   |
| 7      | Explorer workflow         | <code>gemstone-rs-explorer</code>                                                               | <code>tooling/explorer.md</code>                                                                                                                                          | <code>docs/explorer.md</code> , <code>docs/screenshots.md</code>           | <code>python-database</code>          |

| Stream | Feature                  | Rust surface                                                                                          | Examples                                                                                                     | Docs                                                                | gemstone        |
|--------|--------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------|
|        |                          |                                                                                                       |                                                                                                              |                                                                     |                 |
| 8      | VS Code workbench        | <code>vscode-gemstone-rs-workbench</code>                                                             | <code>tooling/vscode-workbench.md</code>                                                                     | <code>docs/vscode-workbench.md</code>                               | gemstone        |
| 9      | Rust web services        | <code>gemstone-rs</code> examples plus checked Axum service and installed Axum scaffold               | <code>http_service</code> , <code>examples/axum-service</code> , <code>examples scaffold axum_service</code> | <code>docs/examples-guide.md</code> , <code>docs/cookbook.md</code> | FastAPI example |
| 10     | Release and verification | <code>scripts</code> , <code>Makefile</code> , GitHub Actions                                         | Release verification commands                                                                                | <code>docs/release-checklist.md</code>                              | PyPI/TVSIX re   |
| 11     | Shared native core       | <code>gemstone-gci</code> , <code>gemstone-rs</code> , future <code>gemstone-py-native</code> wrapper | Shared-core integration plan                                                                                 | <code>docs/shared-core-integration.md</code>                        | gemstone        |

## What This Says Compared With `gemstone-py`

`gemstone-rs` is strongest where Rust is naturally valuable: direct native GemStone access, explicit OOP/value handling, typed generated wrappers, BridgeRoot mapping, and CLI/explorer tooling that can run without Python in the process.

`gemstone-py` remains stronger where Python already has mature product shape: broader web framework adapters, async examples, the database explorer, package extras, and broader release surfaces.

The projects should converge by sharing the Rust native core underneath `gemstone-py-native`, while keeping the Python and Rust APIs idiomatic for their respective users.

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